Carlos Alcala

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EDUCATION

Loyola University, Chicago, IL
B.S. Software Engineering — Expected Graduation: **December 2027**

TECHNICAL SKILLS

Languages: Swift, Java, Python, JavaScript, Racket, C++
Technologies & Tools: Firebase, SQL/F1, gRPC, CoreML, OpenCV, Waymax, Twilio, Xcode, GitHub, IntelliJ, SolidWorks
Skills: Backend Development, Machine Learning, Computer Vision, Databases, Kinematic Modeling, iOS Dev, MVVM arch

PROFESSIONAL EXPERIENCE

Google | Software Engineering Intern, Sunnyvale, CA — May 2025 - August 2025

- Developed an internal analytics dashboard to streamline promotion monitoring, providing teams with live visibility into clicks, views, and audience reach.
- Built asynchronous gRPC integrations through Google's Boq serving infrastructure, retrieving promotion metrics from an F1 SQL database to support live analytics and high-traffic requests.

EveryoneCanCode Apple | App Development Instructor, Chicago, IL — Jun 2024 - Aug 2024, Jun 2026 - Aug 2026

- Led Swift and iOS development instruction for 50+ students across two summer programs, guiding teams through application development, debugging, and testing.
- Mentored four student teams to place in the Top 4 of Apple's citywide app competition, supporting the integration of Firebase, Twilio, HealthKit, and CoreML into their applications.

Apple App Accelerator | Software Engineering Intern, Chicago, IL — June 2023 - December 2023

- Engineered an iOS system using Swift and pitched as 1 of only 4 citywide finalist teams selected from a highly competitive incubator program at the Chicago Downtown Apple Store.
- Optimized real-time hospital alerts by building a robust Firebase database and priority-filtering logic for **Kalcom**, a call-light iOS application to help patients receive prioritized attention efficiently.

PROJECTS & LEADERSHIP

Motorcycle Decision-Making Simulation for Emergency Cases | Personal Project — **In Progress**

- Developed a motorcycle agent in Python using Waymax simulator and Motion Dataset to model motorcycle behavior in traffic scenarios.
- Exploring an intelligent rider-assistance system that uses motorcycle kinematics, braking limits, and acceleration capabilities to improve hazard awareness and support life-saving decisions in potential collision scenarios.

Retargeting Kinematics Hand | Research — Apr 2026 - Jun 2026

- Evaluated an existing robotic hand-retargeting pipeline using **MediaPipe** and **PyRoki**, testing how human hand movements translated into robotic joint configurations.
- Experimented with finger-weighted cost functions in a Levenberg–Marquardt optimization solver to investigate improvements in joint alignment and motion replication.

LIGHT Engineering | Team Lead — Nov 2025 -Current

- **Founded** Loyola University Chicago's first engineering project team, growing it to 30+ members across multiple disciplines.
- Leading the design and development of an electric kart for Purdue's evGrandPrix, guiding multiple teams through CAD, fabrication, integration, and testing.

NU Solar Car Team | Mechanical member — Sept 2025 - Current

- Joined NU Solar Car Team as its only non-Northwestern student, contributing to its solar-powered vehicle.
- Designed brake pedals, suspension components, and rotors in **CAD**, validated designs through Finite Element Analysis, and machined parts in the Ford Engineering Design Lab.

Sheltie - Natural Disaster Model | Personal Project — Nov 2024 - Feb 2025

- Developed a real-time structural hazard detection system by training a CoreML computer vision model on structural fatigue datasets and deploying it in an iOS app for live camera-based risk detection and safety guidance.

AWARDS

Engineering: Congressional App Challenge Winner | Top 4 Apps EveryoneCanCode+Create | National Recognition Program Awardee